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MDCT-Unet: A dual-encoder network combining multi-scale dilated convolutions with Transformer for medical image segmentation

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ABSTRACT

Precise medical image segmentation is crucial in clinical diagnosis and pathological analysis. Most segmentation methods are based on U-shaped convolutional neural networks (U-Net). Although U-Net performs well in medical image segmentation, as a method based on CNN, its main drawback lies in the difficulty of establishing long-range pixels dependencies and has a constrained receptive field, which restricts segmentation accuracy. Many models address this issue by incorporating Transformer models into U-Net architectures to better capture long-range dependencies. However, these methods often suffer from simple feature fusion techniques and limited receptive fields for local features. To address these challenges, we propose a dual-encoder framework, named MDCT-Unet, which combines Swin-Transformer and CNN for enhanced medical image segmentation. This framework introduces a novel dynamic feature fusion module to better integrate of local and global features. By combining channel and spatial attention mechanisms and inducing competition between them, we enhance the coupling of these two types of features, ensuring richer information representation. In addition, to better extract multi-scale local features from medical images, we design a dilated convolution encoder (DCE) as the CNN branch of our model. By incorporating dilated convolutions with varying receptive fields, DCE captures rich local features at multiple scales, thereby enhancing the model's ability to segment challenging regions such as boundaries and small organs. We conducted extensive experiments on four datasets: Synapse, ISIC2018, CHASEDB1, and MMWHS. The experimental results show that our method outperforms most current medical image segmentation methods quantitatively and qualitatively.

1. Introduction

In recent years, deep learning has shown remarkable success in clinical diagnosis and treatment planning. Its powerful capability to learn hierarchical representations from medical data has enabled substantial advancements in tasks such as disease classification [1–3], lesion detection [4–6], and prognosis prediction [7,8]. As a fundamental step in many of these applications, medical image segmentation is essential for computer-assisted diagnosis and therapy, assisting doctors in quantifying diseases, assessing prognoses, and evaluating treatment outcomes. However, manual medical image segmentation is time-consuming and labor-intensive, requiring skilled professionals to invest significant effort. As a result, there has been widespread interest in automated segmentation methods.

With the advancement of deep learning in the field of image processing, including those based on Generative Adversarial Networks (GANs) [9] and convolutional neural networks (CNNs) [10,11]. CNNs with their powerful feature extraction capabilities, dominate many medical image segmentation tasks. Notably, the encoder–decoder architecture, models like FCN [10] and U-Net [11], has demonstrated excellent performance. To address different tasks requirements, researchers have designed a series of U-Net-based variants [12–15]. These models enhance multi-scale feature fusion through skip connections. Despite their success, CNNs face limitations. Since a single convolutional kernel can only focus on a specific region within an image, CNN-based networks are unable to capture global contextual links and build long-range interdependence. [16]. This limitation restricts their performance in tasks requiring precise boundary detection or understanding complex spatial relationships.

Computer vision using Transformer-based networks has been quite successful as of late due to their ability to capture long-range semantic information and global contextual relationships [17]. Examples of such networks include Vision Transformer (ViT) [18], Swin-Transformer [19], and Data-efficient Image Transformer (DeiT) [20].

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As a result, the application of Transformers in medical image segmentation has also garnered increasing attention from researchers. For example, Cao He et al. replaced the traditional CNN modules with Swin-Transformer blocks, creating a U-shaped Transformer model known as Swin-Unet [21]. This model outperformed other models on the Synapse and ACDC datasets. Similarly, Chunfu Chen et al. proposed CrossVIT [22], a dual-branch network with a cross-attention module to integrate multiscale features. Ailiang Lin et al. proposed DS-TransUnet [23], which further enhances feature representation by dividing images into patches of varying sizes. In IIAM [24], the authors generate more reliable predictions by employing a multi-scale attention mechanism and enforcing prediction consistency across decoders of different types. However, despite the excellent performance of these Transformer-based models, they compress images into one-dimensional tokens and calculate attention weights only at the patch level often neglects intra-patch pixel relationships. This limitation can impact their ability to segment blurred boundaries accurately.

To address the limitations of using either a single CNN or a Transformer architecture, researchers have begun to explore hybrid models that combine the use of both, such as TransUnet [25]. As a pioneering application combining Transformer and CNN models in medical image segmentation, the authors combine U-Net's downsampled feature maps with Transformer blocks to establish long-range dependencies. Similarly, FAT-Net [26] integrates a Transformer branch next to the CNN network as the primary encoder, effectively capturing long-range semantic information and global contextual relationships. BGRA-GSA [27] proposed GSAM, which captures features across multiple scales in both dimensions and channels. Despite achieving excellent segmentation results, these methods still face some constraints that limit their performance: (1) The CNN encoders in these models have restricted capability to capture multiscale information, which limits the model's ability to learn local features. (2) The differences in feature information between the two encoders are ignored when fusing features between the CNN and Transformer branches.

In this work, we propose a novel and versatile framework, MDCT-Unet, that effectively combines the strengths of both CNNs and Transformers to address the challenges of medical image segmentation across diverse tasks. MDCT-Unet enhances the receptive field of local features and selectively fuses global and local information by dynamically adjusting their relative importance. As a result, the framework can better differentiate between various tissues in medical images and capture richer boundary details. Specifically, MDCT-Unet employs a dual-encoder architecture comprising a CNN-based branch and a Swin-Transformer-based branch. The CNN branch incorporates a newly designed Dilated Convolution Encoder (DCE) to extract multi-scale, fine-grained local features, while the Swin-Transformer branch focuses on capturing global, coarse-grained context. Furthermore, we introduce a novel feature fusion mechanism that effectively integrates the two types of features while preserving their respective strengths, facilitating better multi-scale representation learning. The fused features are then supplemented into the decoder through skip connections to recover the fine details lost during encoding. We extensively evaluated MDCT-Unet on several public datasets, including Synapse, CHASEDB1, ISIC2018, and MMWHS, covering a wide range of medical image segmentation tasks (organ segmentation, vessel segmentation, and lesion segmentation). The results consistently demonstrate that our method outperforms existing approaches, highlighting its strong generalization capability and broad applicability across different types of medical imaging challenges. The main contributions of this study are summarized as follows:

(1) We propose a dual-encoder framework, MDCT-Unet, which combines Swin-Transformer and our DCE. The DCE captures fine-grained features at various scales, enhancing the model's ability to extract local features at different stages.

(2) We introduce a dynamic feature fusion module (ADFF) that fuses local and global features by combining channel and spatial attention, enhancing feature coupling and preserving rich information.

(3) We conduct extensive experiments on four public datasets — Synapse, CHASEDB1, ISIC2018, and MMWHS — covering diverse medical image segmentation tasks including organ segmentation, vessel segmentation, and lesion segmentation. The results demonstrate that our method consistently achieves highly competitive or superior performance compared to state-of-the-art approaches, highlighting its strong generalization capability and adaptability across a wide range of medical imaging challenges.

2. Related works

2.1. Methods based on CNNs

Before deep learning was widely used for processing medical images, medical image segmentation primarily relied on manual annotation and traditional machine learning methods. In recent years, with the great success of U-Net [11] in the field of image processing, a large number of medical image segmentation models based on U-Net have emerged, achieving excellent performance on various medical image segmentation datasets. For example, ResU-Net [14] enhances the segmentation capability for small blood vessels by introducing weighted attention. U-Net++ [13] improves the model's ability to learn features of different granularities by incorporating dense skip connections. Attention U-Net [28] proposes a novel attention mechanism that dynamically adjusts the weights of feature maps, allowing the model to focus more on regions of interest. Double U-Net [29] introduces a tandem dual encoder–decoder structure by chaining U-Net and incorporating Atrous Spatial Pyramid Pooling (ASPP) [30] to capture multiscale feature information better. Similarly, CE-Net [15] introduces Atrous Convolution and Pyramid Pooling at the bottleneck layer to extract rich feature information at different receptive fields. However, they still struggle to effectively capture global contextual information [31] and their encoder designs lack sufficient consideration for multi-scale feature extraction, which limits their ability to capture local features.

2.2. Methods based on transformers

Transformers [17] were originally designed for natural language processing but have now been widely applied to various image tasks. Their self-attention mechanism enables them to capture global contextual information and extract multiscale features, which is crucial for achieving high-precision medical image segmentation. ViT [18], the first model to apply Transformers to vision tasks. It demonstrated the feasibility of exploring the application of Transformers in image-related tasks. Swin-Transformer [19], building upon ViT, introduced a shifted window-based self-attention mechanism, which captures global contextual information while reducing computational complexity. Building on these works, numerous models that apply Transformers to medical image segmentation have emerged. For example, Swin-Unet [21] integrates Swin-Transformer with a U-Net inspired model, utilizing sliding window attention to effectively capture local and global contextual information. This model has shown outstanding performance in challenging segmentation tasks, such as tumor and organ segmentation. MedT [32], a Transformer-based model specifically developed for the task of medical image segmentation, introduces a gated attention mechanism based on Axial-Deeplab [33]. It also employs multi-scale attention to capture varying organ sizes and shapes, providing reliable segmentation across different modalities and anatomical structures. Although Transformer-based models effectively capture long-range dependencies and global information in images, its operation of dividing the images into different patches may lead to neglecting intra-patch details, thereby hindering the model's ability to accurately segment tissue boundaries.

2.3. Methods based on CNN-Transformer

Recent research has focused on combining CNN and Transformer to use their complementary strengths. These hybrid models have shown significant improvements in segmentation accuracy and robustness. TransUnet [25] is one of the earliest models to propose the combination of Transformer and CNN. It integrates the Transformer's advantage in capturing global features with U-Net's strong capability in extracting fine-grained features through a hybrid encoder design. Similar to TransUnet, models like HiFormer [34] and TransFuse [35] have also been proposed to address medical image segmentation by integrating Transformer and U-Net. They all employ feature fusion modules to merge the local and global feature information obtained from both branches. However, despite their excellent performance on different datasets, these models still have some limitations in their feature fusion strategies. (1) The interaction between local and global features is limited, which is not conducive to the coupling of these two types of features. (2) They apply channel and spatial attention to only one type of feature, resulting in attention weights that lack comprehensive information. To address these issues, our proposed framework, MDCT-Unet, provides a comprehensive, attention-based feature fusion approach. Benefiting from our ADFF module, MDCT-Unet can better induce the coupling of the two types of features by integrating channel and spatial attention and triggering competition between them. This results in attention weights that contain rich spatial and channel information, thereby better combining the advantages of both networks.

3. Methodology

In complex medical image segmentation tasks, different organ structures often involve distinct task requirements. For organs with a large foreground area, the model needs to capture long-range dependencies to fully exploit global contextual information, while for small organs and boundary regions, fine-grained local features are equally critical. To address this challenge, we propose a framework with dual encoders that allows us to effectively capture both local and global information.

The proposed MDCT-Unet framework employs a U-shaped network structure. As shown in Fig. 1, the network design follows an encoder–decoder pattern, with skip connections to recover some of the details lost during downsampling. The framework comprises four main components: the dilated convolution encoder (DCE), the Swin-Transformer encoder, the attention-based dynamic feature fusion module (ADFF), and the decoder. The DCE and the Swin-Transformer encoder operate in parallel, extracting local and global contextual features from the input medical images, respectively. The ADFF module then integrates the extracted local and global features. The fused information is added to the decoder through skip connections. Finally, the CNN-based decoder restores the fused features to the original resolution of the input image, performing iterative upsampling operations until the final map is generated. It is worth noting that during the decoding phase, the loss is calculated by comparing the output of each decoder layer after convolution with the downsampled labels. We will now proceed to introduce each of these four components in detail.

3.1. Dilated convolution encoder

In image segmentation, the size of the receptive field plays a crucial role in determining the segmentation results. Generally, different receptive fields correspond to information at different scales, and such multi-scale information is essential for effectively capturing local features [36]. In CE-Net [15], the authors designed a multi-scale dilated convolution module to capture local features at various scales. Similarly, Double U-Net [29] employs the ASPP module to enhance multi-scale feature extraction from the encoder, thereby improving segmentation performance. Nonetheless, these methods still fall short in adequately addressing multi-scale representation within the encoder design.

To address this limitation, inspired by DAC in CE-Net [15], we propose a novel Dilated Convolution Encoder (DCE), which serves as a CNN branch that extends multi-scale feature extraction beyond the bottleneck layer to the feature extraction stage. This encoder applies dilated convolutions at multiple scales to capture local features with varying receptive fields. It consists of several Dilated Convolution Blocks (DCBs), which enhance the model's ability to represent multi-scale features through cascade branches with different receptive field sizes.

A large receptive field R is one of the key characteristics of dilated convolution. The receptive field of a single dilated convolution can be calculated as follows:

$$R = (K - 1) \times r + 1 \tag{1}$$

where K denotes the kernel size and r is the dilation rate. This equation indicates that the receptive field increases with larger dilation rates, enabling the convolution to capture broader contextual information.

However, the choice of dilation rate plays a crucial role in the effectiveness of dilated convolutions. On the one hand, a large dilation rate significantly enlarges the receptive field but leads to sparse feature sampling, which weakens the model's ability to extract fine-grained details. On the other hand, a very small dilation rate approximates standard convolution, which preserves fine-grained feature extraction but limits receptive field expansion and may result in redundant computations.

To address this trade-off, we design a multi-branch structure with progressively increasing dilation rates to balance receptive field expansion and feature scale, thereby enabling efficient multi-scale feature extraction. Specifically, as illustrated in Fig. 2, the first branch in each DCB employs standard convolution (dilation rate = 1) to preserve fine local details; the second branch adopts dilation rates of 1, 3, moderately expanding the receptive field while maintaining fine features; and the third branch further extends the dilation rates to 1, 3, 5, allowing the model to capture broader contextual information. Finally, we integrate these features through a Concat+Conv operation and employ residual connections to alleviate the vanishing gradient problem. Unlike conventional approaches that restrict multi-scale processing to the bottleneck layer, DCE shifts multi-scale modeling to the feature extraction stage, allowing local structural details and hierarchical contextual information to be integrated earlier in the encoding process. By this strategy, dilated convolutions expand the receptive field, enabling the model to capture multi-scale feature representations while preserving spatial resolution. This allows DCE to simultaneously encode fine-grained boundary information and broader contextual structures, thereby enhancing its adaptability to various medical image segmentation tasks.

Moreover, although progressively increasing dilation rates can enhance multi-scale feature modeling, improper rate increments may cause the gridding effect, which introduces discontinuities in spatial coverage and hinders fine structure representation. To address this, we carefully design the dilation rate progression to enlarge the receptive field while avoiding the gridding effect. Specifically, following the guidance in [37] we enforce the following constraint on the dilation rate sequence across branches:

$$R_x = \max\left[r_x, R_{x+1} - 2r_x,\ R_{x+1} - 2(R_{x+1} - r_x)\right] \tag{2}$$

where r_x denotes the dilation rate of the xth layer and R_{x+1} is the size of the receptive field of the next layer. This formulation ensures a smooth transition in receptive field sizes and prevents overly sparse sampling patterns that may lead to gridding effect.

Through this design, our model can effectively expand its receptive field without suffering from the gridding effect. As illustrated in Fig. 3, compared with conventional convolutional designs, our proposed DCB structure not only significantly enlarges the receptive field but also captures multi-scale information while avoiding gridding effect.

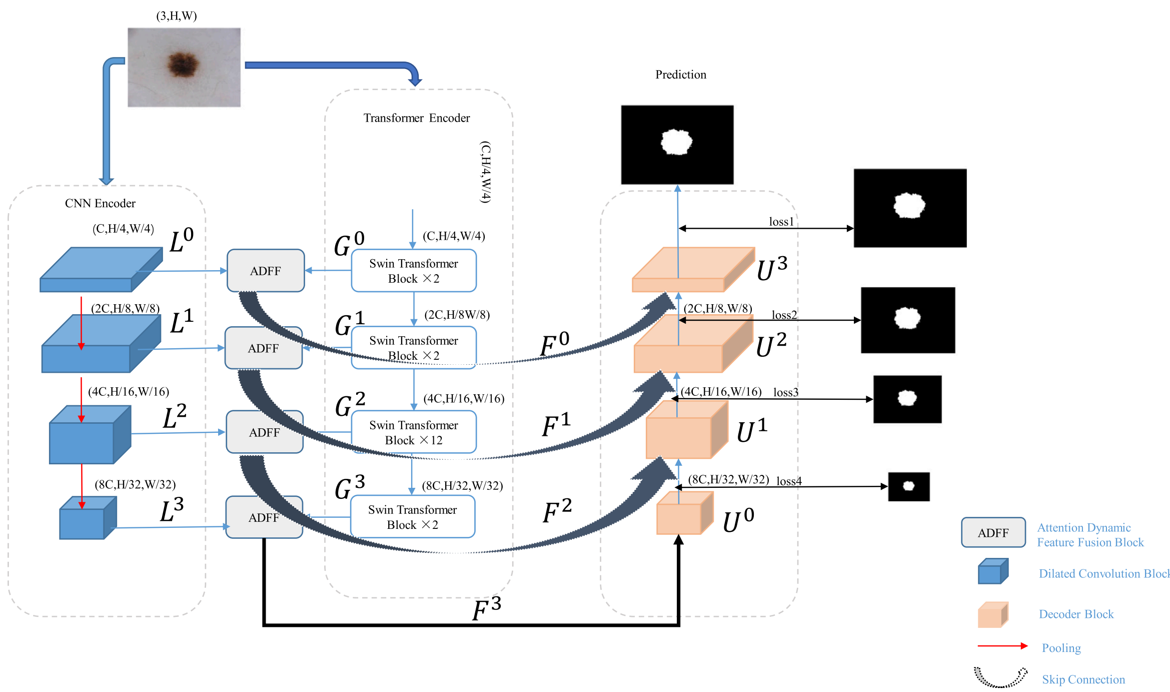


Fig. 1. The overview of MDCT-Unet. L^i and G^i represent the feature maps at each stage after being processed by DCE and Swin-Transformer encoder, respectively, where $i = 0, 1, 2, 3$. F^i represents the result after their fusion. U^i represents the feature maps at each stage after being processed by the decoder. The channel number C, height H and width W of the feature graph of the model after processing at each stage are marked next to each block.

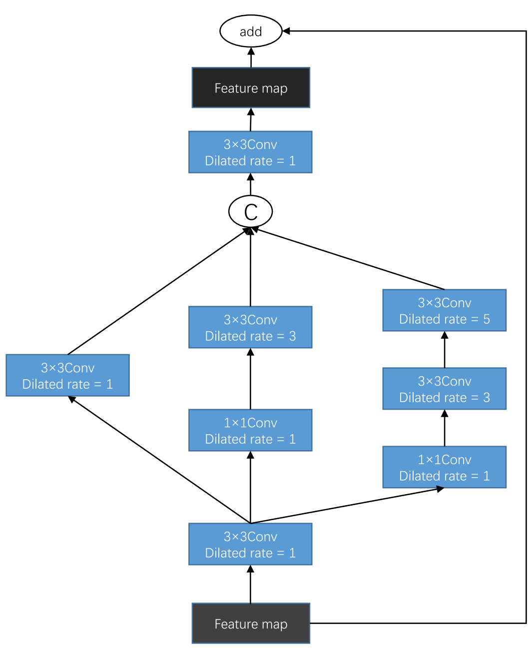


Fig. 2. The overview of proposed DCB. In each branch, the receptive field sizes are 5, 9, and 19, respectively.

3.2. Transformer encoder

To better capture global features from images, we employ the Swin Transformer [19] as an encoder for obtaining global contextual information. Unlike traditional convolutional encoders, Swin Transformer utilizes a hierarchical architecture with shifted windows, which enables efficient modeling of long-range dependencies. In addition, to reduce the model's complexity while maintaining competitive performance, we adopt the Swin-Transformer-Tiny (Swin-T) variant. Swin-T has been widely validated in both natural and medical image analysis tasks, and is known for achieving a good trade-off between model complexity and segmentation accuracy, making it a suitable choice for our framework.as the transformer encoder in our model. The Shifted Window Attention (SW-Attention) mechanism within the Swin-Transformer allows the model to establish direct cross-layer attention connections at different positions, enabling the model to build global relationships across varying resolutions. In addition, the Swin-Transformer achieves global information while reducing computational complexity through the Patch Merging operation, which merges patches at different hierarchical levels. In Fig. 4, we can see two successive Swin-Transformer blocks. These blocks include LayerNorms (LN), W-MSA, SW-MSA, and two Multilayer Perceptrons (MLPs) with two completely linked layers each.

3.3. Feature fusion block

During the integration of features from CNN and Transformer branches, the core challenge lies in effectively merging two structurally distinct types of representations while preserving semantic completeness and consistency. Conventional methods typically apply direct addition of the feature maps followed by decoding, but such simple fusion lacks the capacity to model the semantic gap between local and global features, often leading to suboptimal integration or even semantic conflict.

To address the above issue, we propose an Attention-based Dynamic Feature Fusion (ADFF) module. This module integrates both channel and spatial attention mechanisms to perform attention-based weighting and competitive coupling between local and global features during the fusion process. Such a design facilitates more accurate and consistent feature alignment, mitigates conflicts between complementary information sources, and effectively enhances segmentation performance.

Specifically, we obtain the fused feature F^i, $i = 0, 1, 2, 3$, through the following operations:

$$B^i = Concat(L^i, G^i) \tag{3}$$

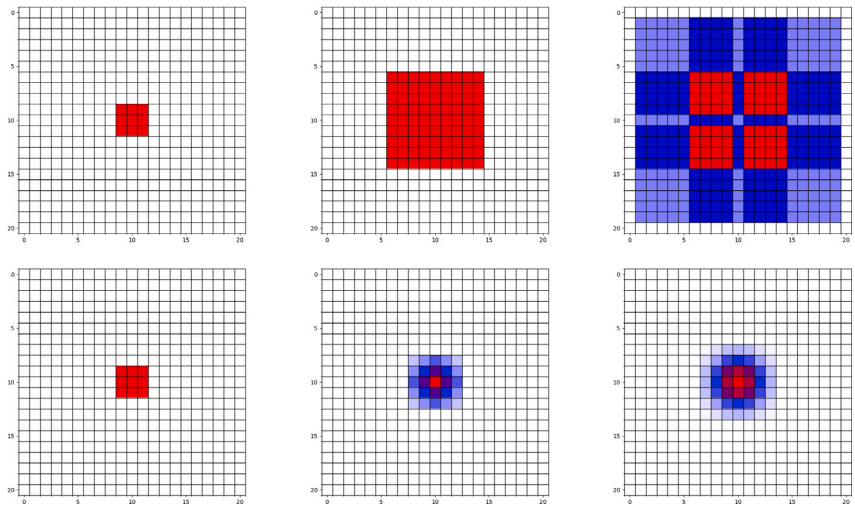


Fig. 3. Visualization of dilated convolution receptive fields. The top row shows the receptive field distribution of three parallel branches in DCB, with dilation rates increasing from left to right; the bottom row displays the receptive fields of standard convolutions for comparison.Red regions indicate high-frequency sampling positions, blue indicates low-frequency sampling positions, and white areas do not participate in feature extraction.

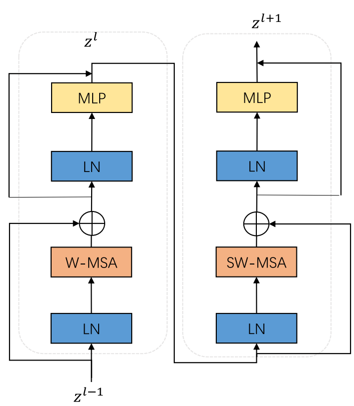


Fig. 4. The Swin-Transformer block.

$$S^i = SpatialAttention(B^i) \quad C^i = ChannelAttention(B^i) \tag{4}$$

$$X^i = Add(S^i, C^i) \tag{5}$$

$$W_1^i, W_2^i = Split(Softmax(X^i)) \tag{6}$$

$$F^i = Add(W_1^i \odot L^i, W_2^i \odot G^i) \tag{7}$$

We define the feature map B^i as the result of concatenating the feature map L^i from the DCE with the feature map G^i from the Swin-Transformer branch. To integrate the feature information from both branches, we pass B^i through a channel attention branch and a spatial attention branch, resulting in channel attention weights C^i and spatial attention weights S^i, respectively. The process is shown in Fig. 5.

In the channel attention branch, we employ the SE-Attention module [38] to recalibrate the channel-wise responses in B^i based on the global context of each input. In the spatial attention branch, we adopt the spatial attention module (SAM) from CBAM [39] to emphasize informative regions and suppress irrelevant spatial locations in B^i, conditioned on the input content. These attention mechanisms allow the model to adaptively adjust the importance of features in both the channel and spatial dimensions. We then obtain an enhanced feature map X^i by adding S^i and C^i through broadcast addition. The resulting X^i integrates dynamically modulated channel and spatial information, making it more responsive to the semantic variations present in different inputs.

Notably, to establish coupling between the local and global branches, we first concatenate their feature maps and apply a shared Softmax function along the channel dimension to generate attention weights. This mechanism inherently introduces competition between local and global features, enabling the model to adaptively allocate attention and emphasize the most discriminative features across different scales, thereby forming a multi-scale attention distribution that captures both local and global semantics. Subsequently, the Split operation assigns the attention weights to L^i and G^i, producing W_1^i and W_2^i, which are then used to modulate the corresponding feature maps via the Hadamard product. Through this mechanism, the network can dynamically learn the relative importance of the CNN and Transformer branches, achieving an adaptive balance between local and global information. Finally, the weighted features are fused by element-wise addition to produce the representation F^i, effectively integrating both local and global contextual information.

3.4. Decoder

Finally, the decoding part is similar to U-Net. Our decoder consists of four upsampling modules. Each upsampling module upsamples the output of the previous layer and integrates it with the fused features from the skip connections before being passed to the next upsampling module. Specifically, each upsampling module comprises an UpSampling operation (Up) with a scale factor of 2, a 3 × 3 convolution (Conv), batch normalization (BN), and the ReLU activation function. The operations within each module are as follows:

$$U^i = ReLU\left(BN\left(Conv\left(Concat\left(Up\left(U^{i-1}\right), F^i\right)\right)\right)\right) \tag{8}$$

$$U^0 = F^3 \tag{9}$$

3.5. Loss functions

To ensure stable performance improvement during training, we adopt a hybrid loss function to measure the discrepancy between the model's prediction and the ground truth. Specifically, the overall loss is composed of two components, as shown in Eqs. (10) and (11), where

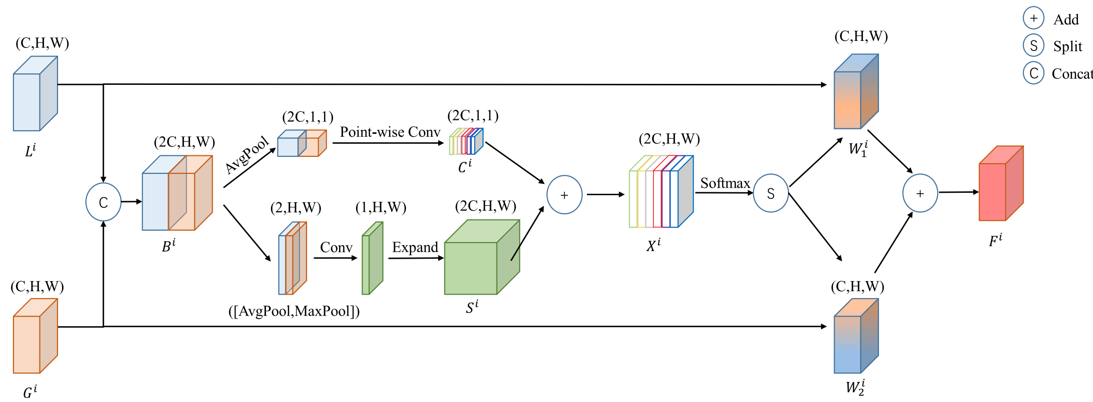


Fig. 5. The schematic diagram of the attention feature iterative fusion block.

P denotes the predicted probability map, and T represents the ground truth.

$$\mathcal{L}_{ce}(P,T) = -\sum_{k=1}^{K} T_k \log(P_k) \tag{10}$$

where K is the number of classes, $T_k \in \{0,1\}$ indicates whether the pixel belongs to class k, and $P_k \in [0,1]$ denotes the predicted probability for class k.

$$\mathcal{L}_{dice}(P,T) = 1 - \frac{2\sum_i P_i T_i}{\sum_i P_i + \sum_i T_i} \tag{11}$$

where i indexes the pixel positions

In medical image segmentation tasks, there often exist large class imbalances due to varying organ sizes. If only $\mathcal{L}_{ce}$ is used as the optimization objective, the model tends to favor majority classes and may ignore small but clinically important regions, especially in multi-class scenarios. Dice Loss $\mathcal{L}_{dice}$ addresses this issue by directly optimizing the spatial overlap between prediction and ground truth, encouraging the model to focus on hard-to-segment small structures. However, Dice Loss alone may lead to unstable gradients during early training, particularly when the initial predictions deviate significantly from the ground truth, resulting in fluctuating optimization behavior.

To balance pixel-wise classification accuracy and structural segmentation quality, we employ a weighted combination of CE and Dice losses. The final loss function is defined as:

$$\mathcal{L}_{total} = 0.5 \cdot \mathcal{L}_{dice} + 0.5 \cdot \mathcal{L}_{ce} \tag{12}$$

Following the settings in [15,40], we set the weights of the two losses to 0.5 each.

4. Experiments

We evaluated the segmentation performance of our method in four common datasets: Synapse multiorgan segmentation, ISIC2018, CHASEDB1, and MMWHS. These evaluations provide insights into how well the model performs across various tasks.

4.1. Datasets

Synapse [41]: This dataset contains 30 cases and 3779 clinical CT images. There are 85 to 198 slices with a 512×512 resolution in every CT volume. The spatial resolution of the voxel is $([0.54 \sim 0.54] \times [0.98 \sim 0.98] \times [2.5 \sim 5.0])\ \mathrm{mm}^3$. Following the same division as in TransUnet, we used 18 cases for training and the remaining 12 cases for validation. The Mean Dice Coefficient (DSC) and the 95% Hausdorff Distance (HD) are employed to evaluate the performance of the model on 8 organs.

ISIC 2018 [42,43]: The dataset is a publicly dataset focused on skin lesion images, which contains 2596 images, each annotated with a resolution of 224 × 320 pixels. We randomly selected 2076 images for training and 520 images for testing. This process was repeated 5 times, and the average value was taken. We used four metrics: DSC, Intersection over Union (IoU), Recall, and Precision, to evaluate the model's performance on the ISIC 2018.

Table 1
Model configuration table. D_e expresses the embedding dimension. #Depth refers to the number of Transformer modules in each layer of the network. #Head represents the number of heads in the multi-head attention modules within each layer. WS represents window size.

CNN encoder	Swin transformer			
	D_e	#Depth	#Head	WS
DCE	96	[2,2,6,2]	[3,6,12,24]	7

CHASEDB1 [44]: This dataset is a vascular segmentation dataset containing 28 color images, each with a resolution of 999 × 960. In this work, we used the first manual annotation. The original dataset does not split the images into training and test sets. Similar to [45], we randomly divided the data into training and evaluation sets, with 21 images used for training and the remaining 7 images used for evaluation. We assessed our results using clDice [46], IoU, and DSC.

MMWHS [47]: A multi-class heart dataset, the MMWHS includes 20 3D MRI images. Myocardium (Myo), ascending aorta (AAo), pulmonary artery (PA), left ventricle (LV), left atrium (LA), right ventricle (RV), right atrium (RA), and myocardium are the seven classes included in the photos. With a ratio of 10:8:2, the dataset is partitioned into three parts: training, testing, and validation. We used DSC and IoU to evaluate the model's performance on the MMWHS.

4.2. Experiment details

In this study, we conducted all experiments based on Python 3.7, PyTorch 1.8.0, and CentOS 7.9. The hyperparameter settings varied across datasets. For the Synapse dataset, we followed the preprocessing steps outlined in [23,34], with an image size of 224 × 224 during training. For the ISIC2018, we resized the images to 224 × 320 and used the same data augmentation techniques as DconnNet [48]. For the CHASEDB1, to maximize the use of the limited data, the image size was adjusted to 960 × 960, and data augmentation was similar to that used in CE-Net [15]. For the MMWHS dataset, our preprocessing operations are limited to standardizing the image data and resizing the images to 224 × 224. During training, we set the batch size to 12 and the maximum number of epochs to 300. All our experiments were conducted using an Nvidia GeForce RTX 3060 GPU. Table 1 depicts the model's specific configurations.

Table 2
Quantitative results on Synapse dataset (DSC%). Best results in **bold**, second-best underlined.

Method	DSC↑	HD↓	Aorta	Gallbladder	L.Kidney	R.Kidney	Liver	Pancreas	Stomach	Spleen
DARR [49]	69.77	–	74.74	53.77	72.31	73.24	94.08	54.18	45.96	89.90
U-Net [11]	76.85	39.70	89.07	69.72	77.77	68.60	93.43	53.98	75.58	86.67
R50-Unet [25]	74.68	36.87	87.74	63.66	80.60	78.19	93.74	56.90	74.16	85.87
Attn-Unet [28]	77.77	36.02	**89.55**	68.88	77.98	71.11	93.57	58.04	75.75	87.30
DeepLabv3+ [50]	77.63	39.95	88.04	66.51	82.76	74.21	91.23	58.32	73.53	87.43
Swin-Unet [21]	79.13	21.55	85.47	66.53	83.28	79.61	94.29	56.58	76.60	90.66
MISSFormer [51]	80.74	19.65	85.31	66.47	83.37	81.65	94.52	63.49	79.63	**91.51**
R50-ViT [18]	71.29	32.87	73.73	55.13	75.80	72.20	91.51	45.99	73.95	81.99
TransUnet [25]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	75.62	85.08
Levit-Unet [52]	78.53	**16.84**	87.33	62.23	84.61	80.25	93.11	59.07	72.76	88.86
MT-Unet [53]	79.19	24.87	87.78	66.90	81.89	79.18	93.72	59.93	74.70	89.47
HiFormer [34]	80.69	19.14	87.03	68.61	84.23	78.37	94.07	60.77	82.03	90.44
PVT-CASCADE [51]	81.06	20.23	83.01	**70.59**	82.23	80.37	94.08	**64.43**	**83.69**	90.10
MDCT-Unet	**81.40**	26.08	88.43	68.10	**88.42**	**83.07**	**95.03**	60.47	78.36	89.28

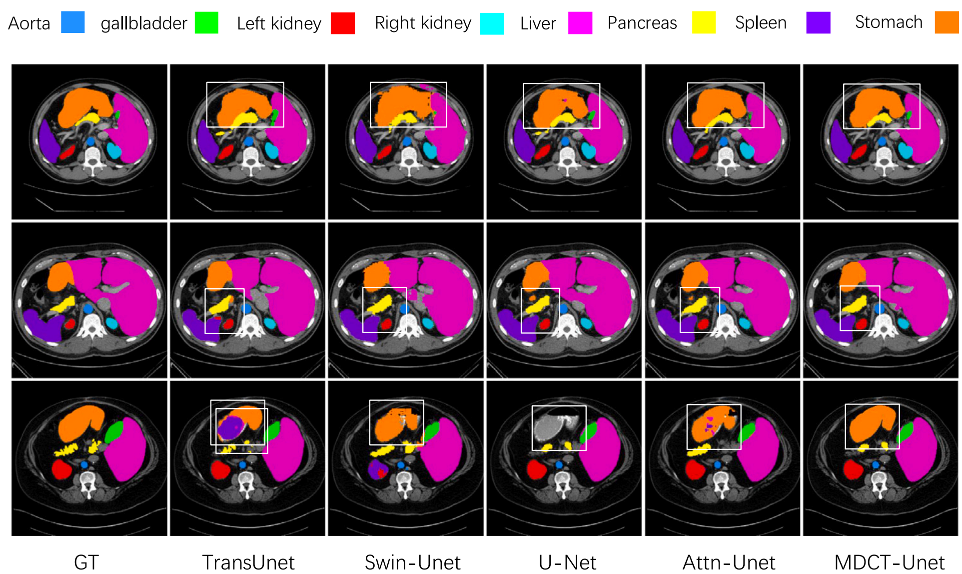


Fig. 6. The Synapse dataset segmentation results visualized, with areas highlighting the advantages of our model marked with white boxes.

4.3. Experiment results

4.3.1. Comparison results on Synapse

We evaluated the performance of our proposed method against 13 state-of-the-art (SOTA) approaches, including CNN-based models (DARR, U-Net, R50-Unet, Attn-Unet, and DeepLabv3+), Transformer-based models (Swin-Unet and MISSFormer), and hybrid CNN-Transformer models (R50-ViT, TransUnet, Levit-Unet, MT-Unet, HiFormer, and PVT-CASCADE). The comparison results are presented in Table 2.

Segmenting four major organs — left kidney, right kidney, spleen, and stomach — presents several challenges. These organs typically have relatively large volumes and are located in close proximity to one another. As a result, the segmentation task demands models that can effectively capture both long-range contextual information and fine-grained boundary features.

Our proposed MDCT-Unet achieves excellent performance in the segmentation of these four organs. Notably, in the segmentation of the left and right kidneys, MDCT-Unet outperforms TransUnet by 6.55% and 6.05%, respectively, achieving the highest performance among all compared methods. For the segmentation of the stomach and spleen, MDCT-Unet achieved Dice scores of 78.36% and 89.28%, respectively, outperforming most existing approaches. These results demonstrate that MDCT-Unet is capable of effectively capturing both detailed local features and long-range dependencies in medical image segmentation tasks.

Among these organs, the liver has the largest area and relatively homogeneous intensity distribution, which makes it more favorable for segmentation compared to smaller or more irregularly shaped organs. Due to its large spatial extent, effective segmentation of the liver relies on the model's ability to capture global contextual information and long-range dependencies. Transformer-based models are particularly well-suited for this purpose. For instance, MISSFormer and Swin-Unet achieved Dice scores of 94.52% and 94.29%, respectively. Classical CNN-based methods such as U-Net and Attn-Unet performed slightly lower, achieving 93.74% and 93.57%, respectively. MDCT-Unet achieved a Dice score of 95.03% for liver segmentation, outperforming MISSFormer by 0.51% and achieving the highest performance among all compared methods. This result demonstrates the strength of MDCT-Unet in modeling global information for large and spatially continuous anatomical structures.

Unlike liver segmentation, the aorta and gallbladder have much smaller areas, and the boundary of the gallbladder often lies close to neighboring anatomical structures. This makes the segmentation task more challenging, as it relies more heavily on precise local feature extraction. Thanks to the strong local feature extraction capabilities of convolutional kernels, CNN-based models such as U-Net and Attn-Unet generally perform better than Transformer-based models in the segmentation of the aorta and gallbladder. For example, U-Net achieved Dice scores of 89.07% and 69.22% for the aorta and gallbladder, respectively, which are significantly higher than the 85.47% and 66.53%

achieved by Swin-Unet. While MDCT-Unet may not surpass pure CNN-based models like U-Net and Attn-Unet in these tasks, it shows notable improvements over Transformer-based models such as MISSFormer and Swin-Unet. Specifically, MDCT-Unet improved the Dice score by 2.96% for the aorta and by 1.57% for the gallbladder compared to Swin-Unet.

Pancreas segmentation is more challenging than that of other organs, primarily due to two reasons: (1) the pancreas often has complex and irregular boundaries, and (2) it is located in close proximity to several other organs, which increases the likelihood of confusion during segmentation. As a result, pancreas segmentation demands a model with strong capability to distinguish between different anatomical structures. Typically, models that combine CNN and Transformer architectures outperform pure CNN-based or Transformer-based models in pancreas segmentation. For example, HiFormer and Levit-Unet achieved Dice scores of 60.77% and 59.07%, respectively, which are significantly higher than those of the CNN-based Attn-Unet and the Transformer-based Swin-Unet. Our proposed MDCT-Unet achieved a Dice score of 60.47% for pancreas segmentation, ranking just behind HiFormer and PVT-CASCADE among the CNN-Transformer hybrid models. Compared to DeepLabv3+ and Swin-Unet, MDCT-Unet achieved improvements of 2.15% and 3.67%, respectively. These results indicate that MDCT-Unet is capable of effectively distinguishing between different organs and accurately segmenting structures with complex boundaries.

Overall, the consistent performance of MDCT-Unet across various organs demonstrates its strong capability to simultaneously capture local texture details via CNN branches and global contextual dependencies via the Transformer encoder. This dual-encoder design enables the model to effectively handle diverse segmentation challenges, making it well-suited for accurate delineation of organs with varying shapes, sizes, and boundary complexities.

The comparison of the segmentation results of MDCT-Unet and other models is shown in Fig. 6. The first row of segmentation images shows that other models, such as TransUnet and Swin-Unet, exhibit boundary confusion when segmenting the stomach, liver, and spleen. In contrast, MDCT-Unet can clearly distinguish organ boundaries and accurately segment smaller structures like the gallbladder and aorta. The second row demonstrates that our model excels at segmenting large organs such as the liver, maintaining the continuity of the segmentation results. In the third row, we can see that U-Net, Swin-Unet, Attn-Unet, and TransUnet all exhibit errors and omissions in stomach segmentation, whereas MDCT-Unet more accurately captures the stomach's boundary contours and internal structures.

4.3.2. Comparison results on ISIC2018

To evaluate the segmentation performance of our proposed method on organs with lesions of different sizes. We evaluated the performance of our model on the ISIC 2018 dataset using DSC, IoU, ACC, and Recall as metrics. The comparison results are summarized in Table 3. Our proposed model outperforms other models on all four metrics. Specifically, MDCT-Unet achieved improvements of 0.45%, 0.67%, and 0.25% in DSC, IoU, and ACC, respectively, compared to the second-best model, HiFormer. In terms of Recall, our model surpassed CPFNet by 0.67%.

Fig. 7 shows various models' segmentation results. For large lesion areas, U-Net and Swin-Unet show boundary inaccuracies, while HiFormer can relatively accurately segment the boundary of the lesion area, but still exhibits some omissions within the segmented area. In contrast, MDCT-Unet effectively segments the lesion area's boundary and internal structure. When dealing with smaller lesion areas, U-Net, Swin-Unet, and HiFormer tend to segment normal tissue incorrectly, while MDCT-Unet can reduce false positives, ensuring precise segmentation.

Table 3
Quantitative analysis results on the ISIC2018 dataset. In each column, the best result is **bold** and the second-best result is <u>underlined</u>.

Networks	DSC	IoU	ACC	Recall
U-Net [11]	86.80	76.68	95.04	88.27
Attn-Unet [28]	89.29	82.22	95.86	89.47
CPFNet [54]	89.63	82.68	96.02	<u>90.62</u>
CE-Net [15]	89.23	82.34	95.76	90.59
Ms RED [55]	89.84	81.79	95.62	90.49
PolypPVT [56]	90.16	83.04	<u>96.36</u>	90.47
TransUnet [25]	89.88	83.00	96.13	90.40
DA-TransUnet [40]	89.92	82.78	96.26	90.28
Swin-Unet [21]	88.57	81.16	95.78	88.26
HiFormer [34]	<u>90.26</u>	<u>83.36</u>	96.35	90.36
MDCT-Unet	**90.71**	**84.03**	**96.60**	**91.29**

Table 4
Quantitative analysis results on the CHASEDB1 dataset. In each column, the best result is **bold**.

Networks	clDice	DSC	IoU
U-Net [11]	77.98	77.95	63.94
R50-Unet [52]	78.62	78.39	64.50
Attn-Unet [28]	81.54	76.60	66.14
CE-Net [15]	81.66	79.88	66.53
DconnNet [48]	82.72	80.43	67.40
Edge-Aware [54]	82.89	80.68	67.64
MDCT-Unet	**83.62**	**82.47**	**68.83**

4.3.3. Comparison results on CHASEDB1

To evaluate the segmentation performance of our proposed method on organs with topological structures, such as blood vessels, we compared it with SOTA methods on the CHASEDB1 dataset. The comparison results are shown in Table 4. We used clDice, DSC, and IoU metrics to assess the method's performance. As shown in Table 4, MDCT-Unet outperformed other methods across all three metrics. Fig. 8 presents the segmentation results generated by each method. The highlighted regions in the images indicate that our model can clearly segment complex vascular structures while maintaining continuity in the segmented vessels.

4.3.4. Comparison results on MMWHS

To further validate the effectiveness of the proposed MDCT-Unet in multi-class tasks, we conducted experiments on the MMWHS dataset. In these experiments, MDCT-Unet was compared with other widely used methods based on CNNs and Transformers. To evaluate the performance of each model, we selected the DSC and IoU as the evaluation metrics. The comparison results are presented in Table 5. As shown in the table, MDCT-Unet outperforms other models on both metrics when tested on the MMWHS dataset. Segmentation results are visualized in Fig. 9. These results demonstrate that MDCT-Unet achieves more precise segmentation, both in delineating boundaries between different organs and in segmenting organs of varying foreground sizes. Combined with the results from the Synapse dataset discussed earlier, these findings demonstrate that the proposed MDCT-Unet exhibits superior performance in multi-class tasks and holds significant promise for practical applications.

4.4. Discussion

These comparative experiments verify that our proposed MDCT-Unet is a medical image segmentation network that integrates global and local information, effectively addressing a wide range of segmentation tasks. The primary contribution of MDCT-Unet is its ability to efficiently fuse global and local features through the ADFF module, enabling the model to learn information at multiple receptive fields and scales. This allows MDCT-Unet to perform exceptionally well in

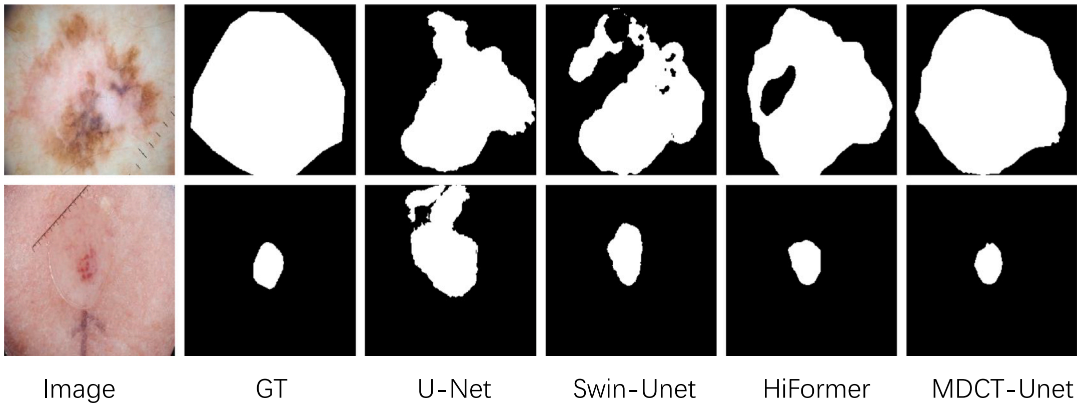


Fig. 7. The ISIC2018 dataset segmentation results visualized.

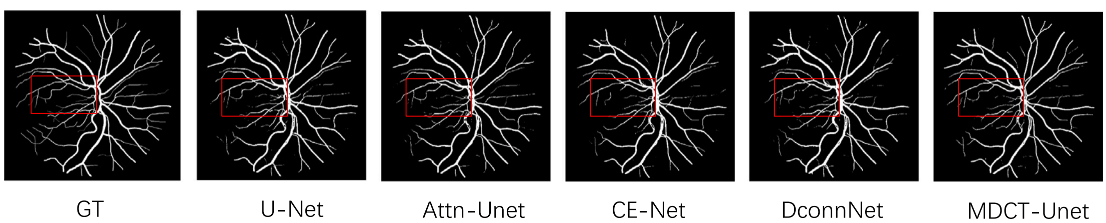


Fig. 8. The CHASEDB1 dataset segmentation results visualized, with areas highlighting the advantages of our model marked with red boxes.

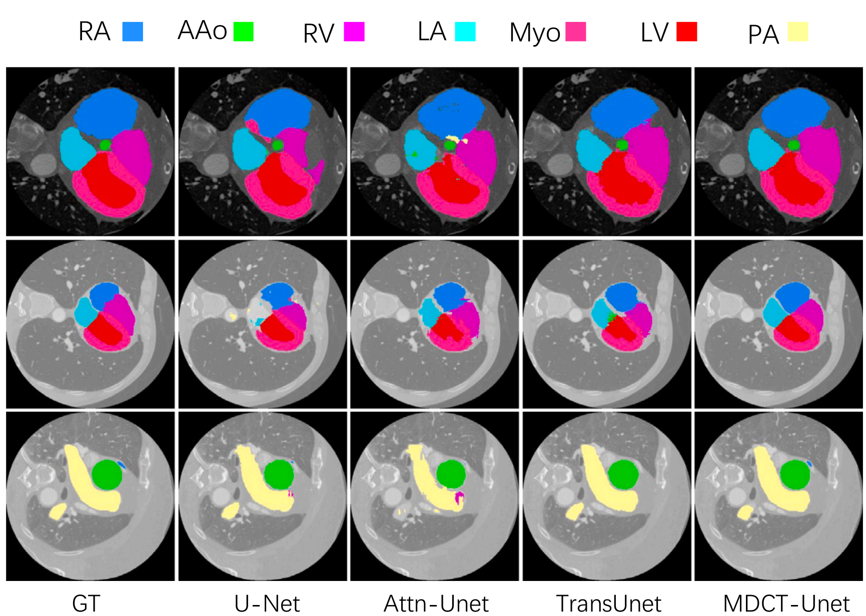


Fig. 9. The MMWHS dataset segmentation results visualized.

Table 5
Quantitative analysis results on the MMWHS dataset. In each column, the best result is **bold**.

Networks	DSC	IoU
U-Net [11]	81.74	70.87
Attn-Unet [28]	81.94	68.87
CE-Net [15]	82.68	70.88
TansUnet [25]	82.94	70.88
Swin-Unet [21]	83.52	74.49
MDCT-Unet	**85.06**	**75.70**

Table 6
Comparison of model parameters. In each column, the best result is **bold**.

Networks	#Parameters(M)	DSC↑	HD↓
R50-Unet [52]	82.46	74.68	36.87
TransUnet [25]	105.28	77.48	31.69
Swin-Unet [21]	**27.17**	79.13	21.55
Levit-Unet [52]	52.17	78.53	**16.84**
HiFormer [34]	29.52	80.69	19.14
MDCT-Unet	51.53	**81.40**	26.08

Table 7
Ablation experiment results with different encoders. In each column, the best result is **bold**.

Networks	Fusion	DSC↑	HD↓
Swin	×	73.70	36.20
R50	×	74.68	36.87
DCE	×	76.94	34.70
Swin+Swin	✓	76.01	32.59
DCE+DCE	✓	78.95	30.84
Swin+R50	✓	78.65	34.93
DCE+R50	✓	78.43	34.58
Swin+DCE	✓	**79.09**	**26.79**

Table 8
Ablation study on different numbers of branches and dilation rates in DCB module. In each column, the best result is **bold**.

Number of branches	1	2	3	4	5
Dilation rates	{1}	{1, 3}	{1, 3, 5}	{1, 3, 5, 7}	{1, 3, 5, 7, 9}
DSC	77.19	79.96	**81.40**	80.63	78.59
HD	32.07	26.59	26.08	**25.32**	26.72

segmenting both large and small foreground regions. Furthermore, the proposed DCE module offers a novel approach to capturing local information by combining atrous convolution kernels with different receptive fields, which enriches the extracted local features and enhances the segmentation performance for small organs and complex organ boundaries.

Experimental results on the Synapse, CHASEDB1, and ISIC2018 datasets demonstrate that MDCT-Unet outperforms widely-used CNN-based and Transformer-based models in multi-class tasks, as well as tasks involving varying foreground sizes and complex topological structures. These results highlight the versatility and robustness of our model across a diverse range of segmentation challenges. Moreover, the segmentation results on the MMWHS dataset, a relatively less-used multi-class dataset, further validate the effectiveness and broad applicability of MDCT-Unet in diverse application scenarios.

Although the superior performance of MDCT-Unet has been demonstrated, model complexity is also a crucial factor in assessing the practicality of a segmentation framework. Therefore, in Table 6, we compare MDCT-Unet with models based on CNNs, Transformers, and hybrid approaches in terms of parameter count and segmentation performance on the Synapse dataset, and analyze the trade-off between accuracy and complexity.

The results show that compared to some traditional models (such as TransUnet, Levit-Unet, and R50-Unet), MDCT-Unet achieves higher performance while maintaining a lower model complexity. Furthermore, when compared to models with parameter counts similar to or lower than ours (such as HiFormer and Swin-Unet), our method consistently outperforms them in at least one evaluation metric. This demonstrates that MDCT-Unet is capable of maintaining a favorable balance between segmentation performance and computational cost.

Finally, from a practical application perspective, the balance that MDCT-Unet achieves between accuracy and complexity is also of great importance for clinical diagnosis. On the one hand, its relatively low parameter count makes it easier to deploy on diagnostic devices to assist physicians in their work. On the other hand, its consistently strong performance across multiple datasets demonstrates good generalization capability, enabling it to adapt to a variety of segmentation scenarios and meet the diverse medical image analysis needs in clinical practice. Therefore, MDCT-Unet ensures segmentation accuracy while maintaining both operational efficiency and adaptability, making it highly promising for real-world applications.

5. Ablation studies

To verify the effectiveness of our proposed MDCT-Unet architecture, we conducted ablation experiments on the Synapse dataset. These experiments aimed to analyze the impact of our modules within the architecture on overall performance.

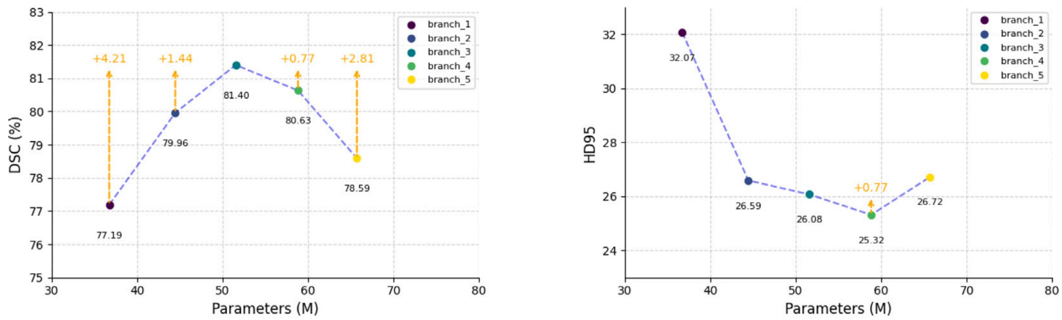
5.1. Efficiency of DCE

To validate the effectiveness of our dual-encoder structure and the DCE, we conducted experiments using different encoder configurations.

In these experiments, "Swin" refers to using the Swin Transformer as the encoder, "DCE" refers to using our proposed Dilated Convolution Encoder and "R50" refers to using ResNet50 [57] as the encoder. All variants use the standard decoder from R50-Unet [25], with skip connections that deliver the output from each downsampling stage of the encoder to the decoder for enhanced feature representation. Building upon this setup, we constructed five dual-encoder variants: Swin+Swin, Swin+R50, DCE+R50, DCE+DCE, and Swin+DCE, where the second encoder is Swin Transformer, R50-Unet encoder, or our proposed DCE, respectively. All feature fusion operations were implemented using concatenation followed by convolution (Concat+Conv). The experimental results are summarized in Table 7.

Firstly, from the comparison among the three single-encoder configurations — DCE, Swin, and R50 — we observe that even without using any fusion strategy, DCE alone achieves superior performance, which confirms the effectiveness of its design. Furthermore, when fixing Swin Transformer as one encoder, the comparisons among Swin+Swin, Swin+R50, and Swin+DCE show that combining Swin with a CNN-based encoder (e.g., DCE) yields better performance than pairing it with another Transformer. This highlights that fusing heterogeneous feature types has a positive impact on segmentation results and improves the model's adaptability across tasks. Similarly, when fixing DCE as one encoder, the results of DCE+Swin, DCE+R50, and DCE+DCE demonstrate that using Swin Transformer as the complementary encoder achieves the best performance, further validating its effectiveness as a global feature extractor.

In addition to comparisons with other commonly used encoders, we conducted an ablation study to further validate the effectiveness of our DCB module by varying the number of branches and the dilation rates of the deepest branch. As shown in Table 8, the model achieves the best DSC performance and the second-best HD when using a three-branch configuration. Fig. 10 illustrates the trade-off between model parameters and performance across different branch settings. Although the four-branch design achieves the best HD, it also introduces a notable increase in parameter count and performs worse than the three-branch configuration in terms of DSC. These results demonstrate that our current design achieves a good balance between model complexity and performance.



(a) (b)

Fig. 10. Effect of different branch designs on model performance.

Table 9
Ablation experiment results with different feature fusion methods. In each column, the highest score is **bold**.

Networks	Fusion	DSC↑	HD↓
Swin	×	73.70	36.20
DCE	×	76.94	34.70
Swin+DCE	Add	77.44	28.20
Swin+DCE	Concat+Conv	79.09	26.79
Swin+DCE(MDCT-Unet)	ADFF	**81.40**	**26.08**

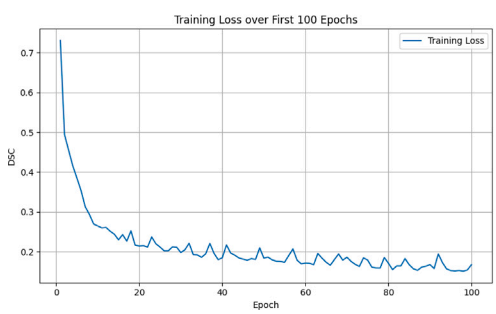


Fig. 11. Training loss over the first 100 epochs.

5.2. Efficiency of ADFF

In this section, we validate the superiority of the proposed Attention-based Dynamic Feature Fusion (ADFF) module compared to other commonly used fusion methods. As in previous experiments, we use “Swin” to denote the Swin Transformer encoder, which is combined with the decoder from R50-Unet. Based on this setup, we compare the ADFF module with two widely adopted fusion strategies: element-wise addition (Add) and concatenation followed by convolution (Concat+Conv). Compared to the simple strategy of Add, our method introduces attention mechanisms that explicitly enhance critical regions in both local and global features, and performs feature fusion through attention-guided weighting. In contrast to the Concat+Conv strategy, which is commonly used in skip connections of U-Net, our ADFF module incorporates a collaborative and competitive mechanism between channel and spatial attention. This enables dynamic adjustment of the importance between the two types of features, thereby improving adaptability to complex structures and multi-scale information. As shown in Table 9, the ADFF module achieves superior performance in terms of both average DSC and HD scores, clearly demonstrating its effectiveness and advancement in feature fusion.

5.3. Analysis of training stability

These experiments were designed not only to analyze the contribution of each proposed module to the overall architecture, but also to comprehensively assess the training stability and reliability of our model. To ensure that the observed performance improvements are not incidental, we present a detailed analysis of training dynamics in this section.

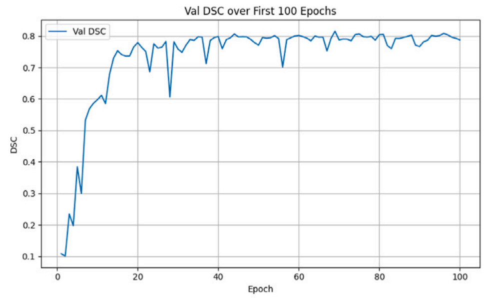
Fig. 11 illustrates the variation of training loss over the first 100 epochs. As shown, the loss exhibits a steadily decreasing trend, indicating a stable convergence process. Although slight fluctuations are occasionally observed, the overall trajectory demonstrates that the optimization process is well-behaved and does not suffer from instability.

Fig. 12 presents the model’s validation performance over the first 100 epochs. Fig. 12(a) shows the evolution of the average DSC over time. It can be seen that the model achieves a consistent and gradual improvement in segmentation accuracy, reflecting reliable generalization capability. Furthermore, Fig. 12(b) shows the per-class DSC across all validation epochs. The results indicate that our model steadily improves performance on all categories, including those with limited spatial extent. This demonstrates not only the model’s effectiveness in handling multi-class segmentation tasks, but also its robustness across anatomically diverse structures.

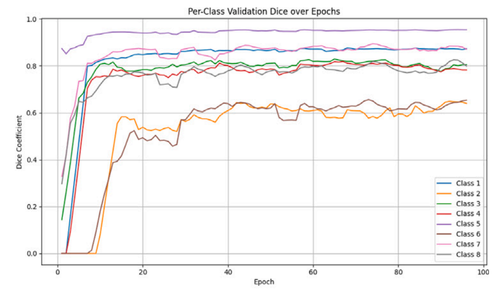
6. Conclusions

In this paper, we propose the MDCT-Unet medical image segmentation network, which effectively combines global and local features through the integration of a Swin Transformer branch and a DCE branch. Our DCE branch differs from conventional CNN-based encoders by employing a multi-branch design with dilated convolutions at different dilation rates, which enables more efficient feature extraction and enhances the capture of fine-grained local details. In addition, compared with prior CNN–Transformer hybrid networks, MDCT-Unet introduces a more efficient feature fusion module, ADFF. By skillfully combining spatial and channel attention mechanisms, the ADFF module effectively fuses multi-scale features obtained from the two encoders. This approach allows the model to integrate both types of features more efficiently, thereby improving its adaptability to a wide range of segmentation tasks. Finally, from a practical application perspective, MDCT-Unet delivers superior performance compared to other models while maintaining a lower parameter count. This demonstrates its efficiency and underscores its potential for deployment in real-world diagnostic scenarios.

Although MDCT-Unet has demonstrated excellent performance across multiple tasks, we still see opportunities to further enhance its application potential. At present, while its parameter count is lower than that of many mainstream models, it remains slightly higher than that of extremely lightweight segmentation networks, indicating room for optimization in scenarios with severely limited computational resources. If the parameter count can be further reduced, this would help lower the difficulty and cost of deploying it as a lightweight model in real-world applications. In addition, MDCT-Unet is currently designed



(a)



(b)

Fig. 12. Validation performance over the first 100 epochs.

specifically for 2D image segmentation and does not yet directly support the processing of volumetric 3D data. We believe that, if the model can be further compressed in size while maintaining accuracy, and its applicability extended to 3D medical image segmentation, its suitability and competitiveness for clinical diagnosis and diverse medical imaging tasks will be further and significantly enhanced.

CRediT authorship contribution statement

Zhengshi Chen: Writing – original draft, Methodology, Conceptualization. **Juanjuan He:** Writing – review & editing, Supervision. **Xiaoming Liu:** Formal analysis, Data curation. **Jinshan Tang:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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